EXT GROUPS OF A PROPER SPACE OVER A NOETHERIAN RING ARE FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1480-1492, immediately following the statement at 1472-1478.

Editorial note. The displayed Ext identity is part of the counted span; the finiteness reading on the Noetherian base is the closing step.

Allowed standard prerequisites (editorial).

- Ext as cohomology of the internal Hom. Cohomology on Sites, Lemma sites-cohomology-lemma-section-RHom-over-U and Section sites-cohomology-section-leray, cited at proof lines 1486-1490. The identity Ext^n(K, L) = H^n(X, RSheafHom(K, L)) = H^n(Spec A, Rf_* RSheafHom(K, L)).
- Coherence of the internal Hom. Lemma lemma-coherent-internal-hom of this chapter, cited at proof lines 1491-1492. RSheafHom(K, L) belongs to D_Coh for K in D^-_Coh and L in D^+_Coh.
- Coherence of bounded-below pushforward. Lemma lemma-direct-image-coherent-bdd-below of this chapter, cited at proof lines 1491-1492. Rf_* of a coherent complex is coherent for f proper.

Source reference. Original lemma 0D0S: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1457–1457. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0D0R: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1437–1437. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08JA: focused-seeds/geometry/S9-perf-ext-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 072X: focused-seeds/geometry/S9-perf-ext-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a Noetherian ring. Let X be a proper algebraic space over A. For L in $D^+_{Coh}(\mathcal{O}_X)$ and K in $D^-_{Coh}(\mathcal{O}_X)$, the A-modules $\mathrm{Ext}^n_{\mathcal{O}_X}(K, L)$ are finite.*

Proof. Recall that

$$\mathrm{Ext}^n_{\mathcal{O}_X}(K, L) = H^n(X, R\mathcal{H}om_{\mathcal{O}_X}(K, L)) = H^n(\mathrm{Spec}(A), Rf_*R\mathcal{H}om_{\mathcal{O}_X}(K, L))$$

see Cohomology on Sites, Lemma 08JA and Cohomology on Sites, Section 072X. Thus the result follows from Lemmas 0D0S and 0D0R. □